

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	18th June 2026
Team ID	
Project Name	MediCareBook (Book a Doctor)
Maximum Marks	4 Marks

Technical Architecture:

The technical stack defines the core architectural layout. A complete stack mapping requires detailing frontend component nodes, application logic controller configurations, and backend database storage services.

Project Implementation: MVC Backend Architecture

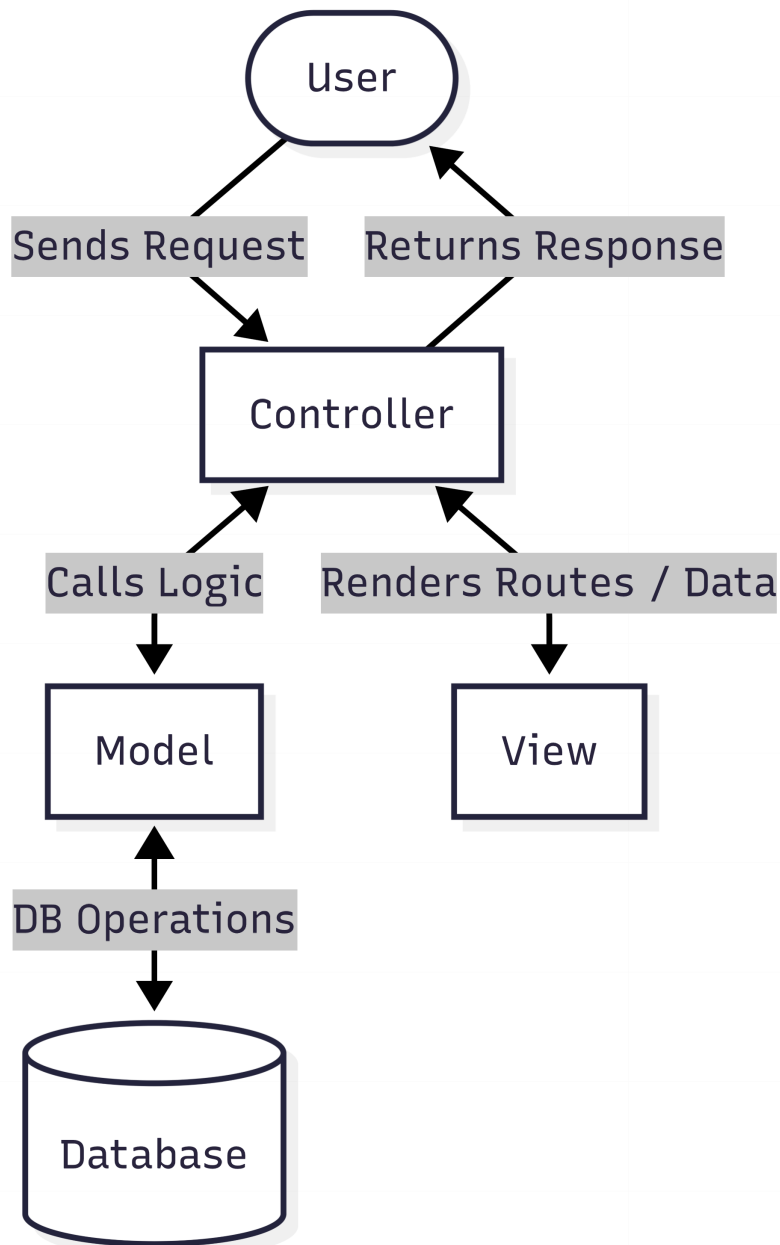


Table-1: Components & Technologies (Book a Doctor Mappings)

S.No	Component	Description	Technology
1	User Interface	Interactive client dashboards and specialized filters.	React.js, Tailwind CSS (Vite)
2	Application Logic	REST routing interfaces and auth verification gates.	Node.js, Express Framework
3	Database	Mongoose model validations and MongoDB document storage.	Mongoose, MongoDB Local Engine
4	Cloud Database	Cloud hosting options for horizontal replica collections.	MongoDB Atlas Clusters

S.No	Component	Description	Technology
5	Security/Cryptography	Cryptographic salting and password hashing on account creation.	BcryptJS

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Stack	Built using free, community-backed libraries, avoiding licensing costs.	React, Express, Node.js
2	Security Implementations	Role permissions middleware block horizontal token escalations.	BcryptJS, Custom RBAC Router Middleware
3	Scalable Architecture	Decoupled frontend/backend layout allows separate endpoint scaling.	Express APIs, Vite Bundling